

## Limits and Continuity

### Complex Analysis

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## Complex Functions

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## Definition

A **complex-valued function** is a function with range in  $\mathbb{C}$  and a **function of complex variable** is a function whose domain is contained in  $\mathbb{C}$ . Furthermore, a **complex function** is a complex-valued function of complex variable.

## Definition

Let  $f$  be a complex function  $f : \Omega \subseteq \mathbb{C} \rightarrow \mathbb{C}$ . If  $\Gamma \subseteq \Omega$ , then the **direct image** of  $\Gamma$  under  $f$  is the subset  $f(\Gamma)$  of  $\mathbb{R}(f)$  given by

$$f(\Gamma) = \{f(z) : z \in \Gamma\}.$$

If  $P$  denotes a property that is meaningful and unambiguous for elements of  $\mathbb{C}$ , we say that property  $P$  holds for  $f(\Gamma)$  if it holds for all  $f(z)$  such that  $z \in \Gamma$ .

## Definition

Let  $\Omega \subseteq \mathbb{C}$ , and let  $z_0$  be a limit point of  $\Omega$ . For a function  $f : \Omega \rightarrow \mathbb{C}$ , a complex number  $L$  is said to be a **limit of  $f$  at  $z_0$**  if, given any  $\varepsilon > 0$ , there exists a  $\delta > 0$  such that if  $z \in \Omega$  and  $0 < |z - z_0| < \delta$ , then  $|f(z) - L| < \varepsilon$ .

## Definition

If  $L$  is a limit of  $f$  at  $z_0$ , then we also say that  $f$  **converges** to  $L$  at  $z_0$ . We often write

$$L = \lim_{z \rightarrow z_0} f(z) \quad \text{or} \quad L = \lim_{z \rightarrow z_0} f.$$

We also say that “ $f(z)$  approaches  $L$  as  $z$  approaches  $z_0$ ”. The symbolism

$$f(z) \rightarrow L \quad \text{as} \quad z \rightarrow z_0$$

is also used sometimes to express the fact that  $f$  has a limit  $L$  at  $z_0$ .

If the limit of  $f$  at  $z_0$  does not exist, we say that  $f$  **diverges** at  $z_0$ .

## Theorem 1.1

*Let  $f : \Omega \rightarrow \mathbb{C}$  and let  $z_0$  be a limit point of  $\Omega$ . Then  $f$  can only have one limit at  $z_0$ .*

## Theorem 1.2

Let  $f : \Omega \rightarrow \mathbb{C}$  and let  $z_0$  be a limit point of  $\Omega$ . Then the following statements are equivalent.

- i)  $\lim_{z \rightarrow z_0} f = L$ .
- ii) Given any  $\varepsilon$ -ball  $B_\varepsilon(L)$  of  $L$ , there exists a  $\delta$ -ball  $B_\delta(z_0)$  of  $z_0$  such that if  $z \neq z_0$  is any point in  $B_\delta(z_0) \cap \Omega$ , then  $f(z)$  belongs to  $B_\varepsilon(L)$ .

### Theorem 1.3 (Sequential Criterion)

Let  $f : \Omega \rightarrow \mathbb{C}$  and let  $z_0$  be a limit point of  $\Omega$ . Then the following are equivalent.

- i)  $\lim_{z \rightarrow z_0} f = L$ .
- ii) For every sequence  $(z_n)$  in  $\Omega$  that converges to  $z_0$  such that  $z_n \neq z_0$  for all  $n \in \mathbb{N}$ , the sequence  $(f(z_n))$  converges to  $L$ .

## Theorem 1.4 (Divergence Criteria)

Let  $\Omega \subseteq \mathbb{C}$ , let  $f : \Omega \rightarrow \mathbb{C}$  and let  $z \in \mathbb{C}$  be a limit point of  $\Omega$ .

- a) If  $w \in \mathbb{C}$ , then  $f$  does **not** have a limit  $w$  at  $z_0$  if and only if there exists a sequence  $(z_n)$  in  $\Omega$  with  $z_n \neq w$  for all  $n \in \mathbb{N}$  such that the sequence  $(z_n)$  converges to  $z_0$  but the sequence  $(f(z_n))$  does **not** converge to  $w$ .
- b) The function  $f$  does **not** have a limit at  $z_0$  if and only if there exists a sequence  $(z_n)$  in  $\Omega$  with  $z_n \neq w$  for all  $n \in \mathbb{N}$  such that the sequence  $(z_n)$  converges to  $z_0$  but the sequence  $(f(z_n))$  does **not** converge in  $\mathbb{R}$ .

## Definition

Let  $\Omega \subseteq \mathbb{C}$ , let  $f : \Omega \rightarrow \mathbb{C}$ , and let  $z_0 \in \mathbb{C}$ . We say that  $f$  is **bounded on a neighbourhood of  $z_0$**  if there exists a  $\delta$ -ball  $B_\delta(z_0)$  of  $z_0$  and a real number  $M > 0$  such that we have  $|f(B_\delta(z_0))| \leq M$ .

## Theorem 1.5

*If  $\Omega \subseteq \mathbb{C}$  and  $f : \Omega \rightarrow \mathbb{C}$  has a limit at  $z_0 \in \mathbb{C}$ , then  $f$  is bounded on some neighborhood of  $z_0$ .*

## Definition

Let  $\Omega \subseteq \mathbb{C}$  and let  $f$  and  $g$  be functions defined on  $\Omega$  to  $\mathbb{C}$ . We define the **sum**  $f + g$ , the **difference**  $f - g$ , and the **product**  $fg$  on  $\Omega$  to  $\mathbb{C}$  to be the functions given by

$$(f + g)(z) := f(z) + g(z), \quad (f - g)(z) := f(z) - g(z),$$

$$(fg)(z) := f(z)g(z),$$

for all  $z \in \Omega$ . Further, if  $\lambda \in \mathbb{C}$ , we define the **multiple**  $\lambda f$  to be the function given by

$$(\lambda f)(z) := \lambda f(z) \quad \text{for all } z \in \Omega.$$

Finally, if  $h(z) \neq 0$  for  $z \in \Omega$ , we define the **quotient**  $f/h$  to be the function given by

$$\left(\frac{f}{h}\right)(z) := \frac{f(z)}{h(z)} \quad \text{for all } z \in \Omega.$$

## Theorem 1.6

Let  $\Omega \subseteq \mathbb{C}$ , let  $f, g$  be functions on  $\Omega$  to  $\mathbb{C}$ , and let  $z_0 \in \mathbb{C}$  be a limit point of  $\Omega$ . Further, let  $\lambda \in \mathbb{C}$ .

i) If  $\lim_{z \rightarrow z_0} f = L$  and  $\lim_{z \rightarrow z_0} g = M$ , then:

$$\lim_{z \rightarrow z_0} (f + g) = L + M, \quad \lim_{z \rightarrow z_0} (f - g) = L - M,$$

$$\lim_{z \rightarrow z_0} (fg) = LM, \quad \lim_{z \rightarrow z_0} \lambda f = \lambda L.$$

ii) If  $h : \Omega \rightarrow \mathbb{C}$ , if  $h(z) \neq 0$  for all  $z \in \Omega$ , and if  $\lim_{z \rightarrow z_0} h = H \neq 0$ , then

$$\lim_{z \rightarrow z_0} \left( \frac{f}{h} \right) = \frac{L}{H}.$$

## Definition

Let  $\Omega \subseteq \mathbb{C}$  and let  $f : \Omega \rightarrow \mathbb{C}$ . If  $z_0 \in \mathbb{C}$  is a limit point of  $\Omega$ , then we say that  $f$  **tends to  $\infty$**  as  $z \rightarrow z_0$ , and write

$$\lim_{z \rightarrow z_0} f = \infty,$$

if for every  $M \in \mathbb{R}$  there is a  $\delta > 0$  such that for all  $z \in \Omega$  with  $0 < |z - z_0| < \delta$ , then  $|f(z)| > M$ .

## Definition

Let  $\Omega \subseteq \mathbb{C}$ , let  $f : \Omega \rightarrow \mathbb{C}$  and let  $D := \mathbb{C} \setminus \overline{B}_r(0)$ . Suppose that  $D \subseteq \Omega$  for some  $r > 0$ . We say that  $w \in \mathbb{C}$  is a **limit of  $f$  as  $z \rightarrow \infty$** , and write

$$\lim_{z \rightarrow \infty} f = w \quad \text{or} \quad \lim_{z \rightarrow \infty} f(z) = w,$$

if given any  $\varepsilon > 0$  there exists  $R > r$  such that for any  $z \in \mathbb{C}$  with  $|z| > R$ , then  $|f(z) - w| < \varepsilon$ .

## Definition

Let  $\Omega \subseteq \mathbb{C}$ , let  $f : \Omega \rightarrow \mathbb{C}$ , and let  $z_0 \in \Omega$ . We say that  $f$  is **continuous at  $z_0$**  if, given any number  $\varepsilon > 0$ , there exists  $\delta > 0$  such that if  $z$  is any point of  $\Omega$  satisfying  $|z - z_0| < \delta$ , then  $|f(z) - f(z_0)| < \varepsilon$ .

If  $f$  fails to be continuous at  $z_0$ , then we say that  $f$  is **discontinuous at  $z_0$** .

**Theorem 1.7**

*A function  $f : \Omega \rightarrow \mathbb{C}$  is continuous at a point  $z_0 \in \Omega$  if and only if given any  $\varepsilon$ -ball  $B_\varepsilon(f(z_0))$  of  $f(z_0)$  there exists a  $\delta$ -ball  $B_\delta(z_0)$  of  $z_0$  such that if  $z$  is any point of  $\Omega \cap B_\delta(z_0)$ , then  $f(z)$  belongs to  $B_\varepsilon(f(z_0))$ , that is,*

$$f(\Omega \cap B_\delta(z_0)) \subseteq B_\varepsilon(f(z_0)).$$

## Definition

Let  $\Omega \subseteq \mathbb{C}$  and let  $f : \Omega \rightarrow \mathbb{C}$ . If  $D$  is a subset of  $\Omega$ , we say that  $f$  is **continuous on the set  $D$**  if  $f$  is continuous at every point of  $D$ . If  $f$  is continuous on  $\Omega$ , then we simply say that  $f$  is **continuous**.

### Theorem 1.8

*Let  $f : \Omega \rightarrow \mathbb{C}$ . Then  $f$  is continuous at  $x_0 + iy_0 = z_0 \in \mathbb{C}$  if and only if  $u(x, y)$  and  $v(x, y)$  are both continuous at the point  $(x_0, y_0)$  (In the two-dimensional real sense).*

### Theorem 1.9

Let  $\Omega \subseteq \mathbb{C}$ , let  $f, g$  be functions on  $\Omega$  to  $\mathbb{C}$ , and let  $\lambda \in \mathbb{C}$ . Suppose that  $z_0 \in \Omega$  and that  $f$  and  $g$  are continuous at  $z_0$ .

- i) Then  $f + g, f - g, fg$  and  $\lambda f$  are continuous at  $z_0$ .
- ii) If  $h : \Omega \rightarrow \mathbb{C}$  is continuous at  $z_0 \in \Omega$  and if  $h(z) \neq 0$  for all  $z \in \Omega$ , then the quotient  $f/h$  is continuous at  $z_0$ .

### Corollary 1.10

Let  $\Omega \subseteq \mathbb{C}$ , let  $f$  and  $g$  be continuous on  $\Omega$  to  $\mathbb{C}$ , and let  $\lambda \in \mathbb{C}$ .

- i) The functions  $f + g$ ,  $f - g$ ,  $fg$  and  $\lambda f$  are continuous on  $\Omega$ .
- ii) If  $h : \Omega \rightarrow \mathbb{C}$  is continuous on  $\Omega$  and  $h(z) \neq 0$  for all  $z \in \Omega$ , then the quotient  $f/h$  is continuous on  $\Omega$ .

### Theorem 1.11

Let  $\Omega \subseteq \mathbb{C}$ , let  $f : \Omega \rightarrow \mathbb{C}$ , and let  $|f|$  be defined by  $|f|(z) := |f(z)|$  for  $z \in \Omega$ .

- i) If  $f$  is continuous at a point  $z_0 \in \Omega$ , then  $|f|$  is continuous at  $z_0$ .
- ii) If  $f$  is continuous on  $\Omega$ , then  $|f|$  is continuous on  $\Omega$ .

**Theorem 1.12**

*Let  $\Omega, \Psi \subseteq \mathbb{C}$  and let  $f : \Omega \rightarrow \mathbb{C}$  and  $g : \Psi \rightarrow \mathbb{C}$  be functions such that  $f(\Omega) \subseteq \Psi$ . If  $f$  is continuous at a point  $z_0 \in \Omega$  and  $g$  is continuous at  $w = f(z_0) \in \Psi$ , then the composition  $g \circ f : \Omega \rightarrow \mathbb{C}$  is continuous at  $z_0$ .*

### Corollary 1.13

*Let  $\Omega, \Psi \subseteq \mathbb{C}$ , let  $f : \Omega \rightarrow \mathbb{C}$  be continuous on  $\Omega$ , and let  $g : \Psi \rightarrow \mathbb{C}$  be continuous on  $\Psi$ . If  $f(\Omega) \subseteq \Psi$ , then the composite function  $g \circ f : \Omega \rightarrow \mathbb{C}$  is continuous on  $\Omega$ .*

## Theorem 1.14

*Let  $f : \Omega \subseteq \mathbb{C} \rightarrow \mathbb{C}$ . Then  $f$  is continuous if and only if  $f^{-1}(\Gamma)$  is open whenever  $\Gamma$  is open.*

### Theorem 1.15

Let  $f : \Omega \subset \mathbb{C} \rightarrow \mathbb{C}$  be continuous and let  $K \subseteq \Omega$ . Then

- i) if  $K$  is compact,  $f(K)$  is compact.
- ii) if  $K$  is connected,  $f(K)$  is connected.

**Theorem 1.16**

*Let  $f : \Omega \subseteq \mathbb{C} \rightarrow \mathbb{R}$ , let  $K \subseteq \Omega$  be compact and let  $f$  be continuous on  $K$ . Then  $f$  assumes an absolute minimum [respectively, maximum] on  $K$ , that is, there exists  $z_\star \in K$  [respectively,  $z^\star \in K$ ] such that for all  $z \in K$  we have  $f(z_\star) \leq f(z)$  [respectively,  $f(z) \leq f(z^\star)$ ].*